



Role of the intestinal mucosa in acute gastrointestinal GVHD

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Intestinal graft-versus-host disease (GVHD) remains a significant obstacle to the success of allogeneic hematopoietic cell transplantation. The intestinal mucosa comprises the inner lining of the intestinal tract and maintains close proximity with commensal microbes that reside within the intestinal lumen. Recent advances have significantly improved our understanding of the interactions between the intestinal mucosa and the enteric microbiota. Changes in host mucosal tissue and commensals posttransplant have been actively investigated, and provocative insights into mucosal immunity and the enteric microbiota are now being translated into clinical trials of novel approaches for preventing and treating acute GVHD. In this review, we summarize recent findings related to aspects of the intestinal mucosa during acute GVHD.

Learning Objectives

- Recall the elements and functions of the intestinal mucosa and intestinal microbiota
- Describe changes that occur in the intestinal microbiota with GVHD
- Describe changes that occur in the intestinal mucosa with GVHD

Introduction

Allogeneic hematopoietic cell transplantation (HCT) is a potentially curative treatment of many benign and malignant hematological diseases but is also associated with significant toxicities. One of the major limitations to allogeneic HCT is graft-versus-host disease (GVHD). The acute form of GVHD, occurring in approximately 50% of patients, depending on the series studied, leads to potentially life-threatening dysfunction of the gut, liver, skin, and hematopoietic organs. The chronic form, which occurs in up to 40% of patients, can be severely debilitating because of a diverse array of pathologies including sclerosis, fibrosis, and exocrine gland dysfunction. Development of either acute or chronic GVHD is the second leading cause of death (after malignant relapse) in allogeneic HCT recipients.

In the 1970s, researchers evaluated allogeneic HCT in mice that were maintained from birth in isolators devoid of microbes¹ or that were receiving gut-decontaminating antibiotics,² and found that both interventions resulted in acute GVHD that was very mild. This indicated an important contributory role of the microbial flora to the development of acute GVHD. The intestinal tract harbors the greatest density of commensal microbes and is also one of the most important target organs

of acute GVHD. Interactions between the allogeneic HCT recipient's intestinal epithelium, stroma, and immune cells with the luminal commensals may thus play a major role in the pathophysiology of acute GVHD. However, the specific nature of these interactions and the best approaches to target this pathophysiology for prevention or treatment of GVHD have not been fully elucidated. In recent years, an increased focus on examining aspects of intestinal mucosal biology³ and advances in approaches to characterize complex mixtures of microorganisms using nucleotide sequencing⁴ have resulted in a better understanding of how this interplay participates in the development and maintenance of GVHD. In this review, we summarize recent efforts that have examined the role of the intestinal mucosa in the setting of acute GVHD.

Structure of the intestine

The intestinal mucosa is the innermost of 4 concentric histological layers found throughout most of the intestinal tract and is followed by the submucosa, the muscularis externa, and the serosa.⁵ The mucosa itself is further subdivided into 3 layers. The epithelium is a single-cell layer lining the interior lumen of the gastrointestinal tract. Immediately adjacent to the epithelial layer is the lamina propria, an interstitial tissue with a rich vascular and lymphatic network and abundant leukocytes. The third is the muscularis mucosae, which is composed of smooth muscle fibers.

Within the epithelium there are many individual cell types, each with its own specialized functions, including nutrient absorption and barrier function, production of mucus, production of antimicrobial molecules, production of growth factors, and cellular regeneration.⁵ Table 1 summarizes the functions of several important intestinal epithelial cell populations. Intestinal epithelial cells (IECs) and the tight junctions between them are critical components of the intestinal barrier and

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Table 1. Selected epithelial cell populations and relationship to allogeneic HCT

Cell type	Homeostatic functions	Sensitivity to damage from conditioning	Reduced in acute GVHD	Effect on acute GVHD
Intestinal epithelial cells, generally, and enterocytes in particular	Barrier function, absorb nutrients, secrete antimicrobial peptides, sense pathogen-associated molecular patterns, and modulate immune cells ⁶⁰	Yes	Yes	Loss of barrier function contributes to acute GVHD ¹¹⁹
Goblet cells	Secrete mucus, transfer antigen to dendritic cells ¹²⁰	Complicated and dynamic response ¹²¹⁻¹²³	Decreased goblet cell frequency in experimental GVHD ¹²⁴	May help prevent GVHD by contributing to barrier function. ⁹⁶ <i>FUT2</i> genotype associated with GVH severity. ¹¹²
Paneth cells	Secrete antimicrobial molecules and provide growth factors to ISCs ⁶	Function is modulated (eg, production of α -defensins) ^{65,125}	Reduced in mouse models and human GVHD ^{57,64-66}	Has not been directly assessed, but genetic mouse studies suggest Paneth cells might dampen acute GVHD ¹²⁶
ISC	Maintain epithelium	Yes, but progenitors are thought to be more radiosensitive ¹²⁷	Reduced in mouse models of acute GVHD ⁵⁵⁻⁵⁷	Supporting ISCs reduces acute GVHD ^{55,57}
Tuft cell (a type of intestinal epithelial cell)	Regulate type 2 innate lymphoid cells through production of IL-25 ¹²¹	Unknown	Unknown	Unknown

are primarily responsible for nutrient absorption. Goblet cells in the epithelial layer secrete mucus that forms an additional barrier of protection. Paneth cells provide supportive functions, including the secretion of antimicrobial peptides and production of growth factors for epithelial stem cells.⁶

Equally important to the intestinal mucosa are immune cells with multiple functions, including monitoring for pathogens and maintaining immune tolerance to food and commensal antigens.⁷⁻¹⁰ These immune cells are generally located within 1 of 3 compartments: the epithelium (termed intraepithelial lymphocytes), the lamina propria (termed lamina propria lymphocytes), and specialized intestine-associated lymphoid structures, which include isolated lymphoid follicles, aggregated lymphoid follicles (named Peyer's patches), and mesenteric lymph nodes.

Community organization of intestinal commensal microorganisms

The human intestinal tracts harbors an estimated 10 trillion bacteria of roughly 1000 species. Approximately 15 000 distinct species of bacteria have been identified in intestinal samples across human populations.¹¹ A simplified guide to the taxonomy of common intestinal bacterial commensals is provided in Table 2. In addition to bacteria, the intestinal tract can also house commensal microbes from the fungi kingdom,¹² as well as a variety of viruses with tropism for commensal bacteria¹³ and host tissues.¹⁴

Interactions between the intestinal mucosa and commensal microorganisms

Underdevelopment of the immune system in germ-free animals

With the advent of techniques to generate and maintain germ-free rodents developed in the 1940s,¹⁵ it became possible to examine how the microbiota contributes to development of the immune system. In these germ-free animals, the development of systemic immune organs such as

the spleen and thymus is not obviously perturbed,¹⁶ although changes in neutrophil function, antigen presentation, and natural killer cell licensing have been observed.^{17,18} In contrast, major changes are seen in intestinal mucosal immunity. In particular, gut-associated lymphoid tissues including isolated lymphoid follicles,^{19,20} Peyer's patches, and mesenteric lymph nodes are all severely underdeveloped.²¹

Improved intestinal barrier function in response to microbial colonization

One response of the intestinal mucosa to colonization by commensal bacteria is to keep the intestinal microbiota safely at bay. In an antigen-specific manner, B cells mature into plasma cells that produce immunoglobulin A (IgA), which is secreted into the gut lumen.²² Epithelial cells can detect bacterial products via innate *MyD88*-dependent signaling receptors and, as a result, generate antimicrobial peptides such as regenerating islet-derived protein 3- γ (Reg3 γ).²³ These molecules have local inhibitory or bactericidal properties.²⁴ Finally, goblet cell-derived mucus generates a difficult-to-penetrate barrier, especially in the colon, to separate intestinal commensals from the epithelium.²⁵

Maintaining tolerance to intestinal antigens

Immune responses in the intestinal tract are uniquely tempered to help prevent development of autoimmunity or allergies while still maintaining the ability to protect against potential pathogens.²⁶ This is critically important because the intestinal lumen is rich in foreign antigens derived from nonpathogenic sources, including intestinal commensals and food. Perhaps the best-studied mediators of immune tolerance to luminal antigens are intestinal regulatory T cells.²⁷ These are present at very low levels in the colons of germ-free mice but can be induced by introduction of bacterial members of the class Clostridia²⁸ and genus *Bacteroides*.²⁹ Interestingly, regulatory T cells in the small intestine are found at approximately normal frequencies in germ-free mice, and appear to be maintained by the presence of dietary

Table 2. Selected common members of the human intestinal microbiota

Phylum	Class	Order	Genus	Gram	Anaerobe
Firmicutes	Bacilli	Bacillales	<i>Gemella</i>	+	Facultative
			<i>Staphylococcus</i>		
		Lactobacillales	<i>Enterococcus</i>		
			<i>Lactobacillus</i>		
	Clostridia	Clostridiales	<i>Streptococcus</i>	+	Obligate
			<i>Blautia</i>		
			<i>Clostridium</i>		
			<i>Eubacterium</i>		
			<i>Faecalibacterium</i>		
			<i>Ruminococcus</i>		
	Erysipelotrichia	Erysipelotrichiales	<i>Erysipelatoclostridium</i> <i>Holdemania</i>	+	Obligate
	Negativicutes	Acidaminococcales	<i>Acidaminococcus</i>	–	Obligate
			<i>Phascolarctobacterium</i>		
		Veillonellales	<i>Megasphaera</i> <i>Veillonella</i>		
Bacteroidetes	Bacteroidia	Bacteroidales	<i>Bacteroides</i> <i>Prevotella</i>	–	Obligate
Proteobacteria	Gammaproteobacteria	Enterobacteriales	<i>Escherichia</i> <i>Klebsiella</i>	–	Facultative
Actinobacteria	Actinobacteria	Actinomycetales	<i>Actinomyces</i>	+	Facultative
		Bifidobacteriales	<i>Bifidobacterium</i>		
Verrucomicrobia	Verrucomicrobiae	Verrucomicrobiales	<i>Akkermansia</i>	–	Obligate

antigens.²⁷ The T-cell receptors of colonic regulatory T cells have been demonstrated to recognize antigens derived from specific commensal bacteria.³⁰ *RORγt*-expressing regulatory T cells in particular play a variety of roles in preventing immune responses against non-pathogenic intestinal antigens, ranging from regulation of T follicular helper cells³¹ to prevention of Th2-mediated inflammation³² and Th1/Th17-mediated inflammation.³³

In addition to regulatory T cells, another immune cell population that has been found to be critical for intestinal mucosal homeostasis is known as innate lymphoid cells (ILCs). ILCs are a recently described group of lymphoid-lineage immune cells that function in an innate fashion without adaptive immune receptors or antigen-specific responses.³⁴ Group 3 ILCs (ILC3s) are tissue-resident cells that produce Th17-family cytokines including interleukin 22 (IL-22); they are most abundant within the intestinal tract and have an important role in maintaining intestinal barrier function.³⁵ In addition, intestinal ILC3s have been shown to express major histocompatibility complex (MHC) class II and are thought to promote tolerance to commensal bacteria, prevent hyperactivation of T cells during enteric infection, and limit intestinal inflammation by presenting commensal antigens to T cells without providing costimulation.^{36–38}

Changes to the intestinal microbiota early in life may have long-term deleterious effects

On birth, the neonate quickly transitions from a nearly sterile environment into a microbe-rich world, and develops, over the course of weeks to months, its own intestinal microbiota.^{39,40} Clinically, microbiota perturbations during development have been associated with long-term immune consequences. Perhaps the 2 most common causes of an altered microbiota in human children include cesarian delivery and antibiotic exposure early in life. The mode of delivery at birth has been found to have a profound effect on the initial

microbiota of newborns,⁴¹ with infants delivered by cesarean delivery displaying an altered microbiota composition consisting largely of skin, rather than vaginal flora. Cesarean delivery, in turn, has in epidemiological studies been associated with allergic and inflammatory conditions.⁴² Similarly, early-life exposure to antibiotics has been associated with subsequent development of allergic conditions,⁴³ inflammatory bowel disease,⁴⁴ and metabolic disorders such as obesity, which may be immune-related.⁴⁵ Potential prevention of late effects in individuals who experience microbiota perturbations with bacteriotherapy is now being examined. A pilot study showed that the intestinal microbiota of newborns delivered by cesarean delivery could be enriched for vaginal bacteria by exposure to maternal vaginal fluids at birth.⁴⁶ Furthermore, mouse models of asthma⁴⁷ and food allergies⁴⁸ indicate that introducing cocktails of intestinal commensal bacteria may be clinically beneficial in these and other conditions. Whether the development of a new immune system within the allogeneic HCT recipient could have parallels to immune development in the neonate is an intriguing concept that has yet to be well-explored.

Components and function of the intestinal epithelium in the setting of GVHD

The effects of conditioning and GVHD on various IEC subsets has been examined predominantly in murine models; a summary of the current findings is provided in Table 1. IECs are connected by tight junctions that establish a physical barrier between luminal contents and the intestinal lamina propria.⁴⁹ Structurally, the surface epithelium of the small intestine forms small finger-like projections termed villi, the interiors of which are supplied by the rich vascular and lymphatic network of the lamina propria. The mature enterocytes of the villous epithelium are supplied by epithelial stem and progenitor cells located within adjacent invaginations of the mucosal surface, termed crypts of Lieberkühn. The intestinal stem cells (ISCs) are located at the crypt base and are capable of generating all cell

types of the intestinal epithelium. They are thus critical both for intestinal homeostasis and for regeneration after injury. Recent advances in the identification of specific molecular markers of ISCs (such as *Olfm4* and *Lgr5*) and techniques for their examination *ex vivo*⁵⁰⁻⁵² have facilitated examination of ISC biology, including in the setting of GVHD.

GVHD has long been suspected of damaging the ISC compartment,^{53,54} but clear data in support of this have been lacking until recently. Immunohistochemistry in mouse small intestine after MHC-mismatched experimental HCT demonstrated a loss of *Olfm4*⁺ ISCs after pre-transplant irradiation and in recipients of allogeneic T cells.⁵⁵ Likewise, a loss of *Lgr5*⁺ ISCs was demonstrated using transgenic *Lgr5* reporter mice in a MHC-matched (minor histocompatibility-antigen mismatched) allogeneic HCT model; this was confirmed by histologic evaluation of ISCs, as determined by crypt-base columnar cell morphology and location in crypts of wild-type mice with GVHD.⁵⁶ Moreover, it was shown that mouse ISCs can be protected from radiation injury and subsequent GVHD by systemic pretreatment with *R-spondin-1*, an ISC growth factor and agonist of canonical Wnt signaling.⁵⁵ Taken together, these studies indicate that injury to ISCs may be an important component of intestinal GVHD pathophysiology, and that strategies to preserve ISCs may be beneficial.

Once damage has occurred in the intestines, an important component of the recovery process is regeneration of the damaged tissue. Recipient-derived IL-22 has been shown to limit the severity of intestinal pathology resulting from GVHD.⁵⁶ However, this protective IL-22 signaling may be lost in intestinal GVHD as a result of the elimination of intestinal IL-22-producing ILC3s.⁵⁶ With the hypothesis of restoring this IL-22-dependent regenerative signal, treatment with recombinant IL-22 starting 1 week after experimental MHC-matched allogeneic HCT was found to improve *Lgr5*⁺ ISC recovery, intestinal GVHD histopathology, and overall survival.⁵⁷ Restoration of this IL-22 signal may also be relevant clinically, where ILC3 deficiency after chemotherapy for initial treatment of acute myeloid leukemia was associated with increased risk of developing GVHD after allogeneic HCT.⁵⁸ Administration of IL-22 along with systemic corticosteroids is now being investigated for initial treatment of lower intestinal GVHD after allogeneic HCT (www.clinicaltrials.gov: NCT02406651). Interestingly, in spite of the possible reduction in ISCs, it has been observed that IECs from patients with acute intestinal GVHD have an increased proliferation index, as assessed by Ki67 staining and reduced telomere length.⁵⁹ This IEC proliferation may represent a response to intestinal damage, and as each cell division produces an incremental reduction in telomere length, it is possible that intestinal recovery may be limited by replicative exhaustion; alternatively, telomerase expression may be reduced.

In addition to ISCs, small intestinal crypts also contain specialized Paneth cells. The Paneth cells sit at the crypt bases in close proximity to ISCs and produce ISC growth factors.⁶ In addition, Paneth cells contain large numbers of secretory granules that store a variety of potent antimicrobial molecules with activity against intestinal bacteria.⁶⁰ Interestingly, an antimicrobial molecule produced by enterocytes, *Reg3α*, was identified in an unbiased screen for clinical plasma biomarkers of intestinal GVHD⁶¹ and was reproducibly found to be predictive of GVHD prognosis when combined with 2 additional plasma biomarkers (*TNFR1* and ST2) in a recent large multicenter study.⁶² Why plasma levels of *Reg3α* correlate with GVHD severity remains an open question, but it may

either reflect epithelial cell damage or be a physiological response to intestinal injury and compromised barrier function. Although production of *Reg3α* and its murine homolog *Reg3γ* have previously been attributed to Paneth cells, it is clear that other non-Paneth-cell IECs produce *Reg3γ* after experimental HCT in mice.^{57,63} This potentially explains how *Reg3* levels can be elevated despite the loss of Paneth cells that has been observed in the settings of experimental^{57,64,65} and clinical⁶⁶ GVHD. Importantly, loss of Paneth cells was found to be a poor prognostic finding in clinical cases of intestinal GVHD.⁶⁶ In addition to GVHD, reductions in Paneth cell frequency have been observed in mouse models in response to treatment with interferon- γ ,^{67,68} a cytokine with a heavily investigated complex pathophysiology in the setting of allogeneic HCT.⁶⁹⁻⁷¹ In contrast to Paneth cell deficiency, the cytokine IL-33 was recently reported to promote Paneth cell differentiation.⁷² IL-33 and its receptor ST2 appear to both play important roles in several aspects of GVHD pathophysiology that are highly context-dependent.⁷³⁻⁷⁵

Components and functions of the intestinal immune system in the setting of acute GVHD

Several arms of the immune system play roles in modulating intestinal inflammation during acute GVHD. Neutrophil infiltration in the small intestine in response to translocation of commensal bacteria or their products has been reported to contribute to development of GVHD via production of reactive oxygen species, which leads to aggravated tissue damage.⁷⁶ As noted earlier, IL-22-producing ILC3s have been shown to be severely reduced in the setting of GVHD.⁵⁶ Antigen-presenting cells are another immune population that contributes to intestinal GVHD severity by directing T cells to differentiate and become activated.⁷⁷ Dendritic cells from mesenteric lymph nodes have been shown to induce T-cell homing to the intestines and focal intestinal damage.^{78,79} Another myeloid population, intestinal monocytes, was found in the setting of GVHD to promote Th17 differentiation,⁸⁰ which is important for recruitment of neutrophils and macrophages to sites of inflammation.

Several groups have found that Th17 differentiation may be an important contributor to intestinal GVHD.⁸¹ More recently, analysis of target organ tissues at the time of GVHD diagnosis also indicated that Th17 differentiation and *STAT3* signaling are prominent in cases of severe GVHD.^{82,83} The cytokine IL-6, which acts upstream to induce Th17 differentiation, has also been identified as a potential target to prevent GVHD in murine models.^{84,85} Tocilizumab, a monoclonal antibody antagonist of the IL-6 receptor that is approved for the treatment of rheumatoid arthritis, produced promising results as a GVHD-prophylaxis agent in a single-arm phase 1/2 study.⁸⁶ Additional clinical trials with tocilizumab in allogeneic HCT recipients are ongoing (www.clinicaltrials.gov: NCT02206035, NCT02057770, NCT02447055). Interestingly, in addition to the T-helper cells described earlier, effector CD8 cells can also undergo Th17-like differentiation and contribute to GVHD,⁸⁷ and IL-6 also is a key agent in directing this type of effector differentiation.⁸⁸

In addition to T cells, the B-cell group of the mucosal immune system is also perturbed in the setting of GVHD. Reduced IgA levels in the intestinal lumen of mice can be seen after HCT⁶⁴ and in the serum of patients during the first 6 months after HCT. Those who develop acute or chronic GVHD can remain chronically IgA-deficient.⁸⁹ Whether decreases in IgA levels or genetic IgA deficiency of the donor or host can contribute to GVHD pathophysiology has not, to our knowledge, been thoroughly investigated.

Commensal aspects of the intestinal mucosa in GVHD

Changes to intestinal bacteria in the setting of GVHD

Development of GVHD can lead to significant changes in intestinal microbiota composition. GVHD in both mice^{11,30,35} and humans^{30,40,41} is associated with loss of intestinal bacterial diversity and an expansion of 2 groups of bacteria: Gram-negative Enterobacteriales (including *Escherichia*, *Klebsiella*, and *Enterobacter* species) and Gram-positive Lactobacillales (including *Lactobacillus*, *Enterococcus*, and *Streptococcus* species). This expansion is coupled with loss of obligately anaerobic Gram-positive bacteria including Clostridia.^{64-65,90} Reasons why acute GVHD leads to changes in bacterial flora composition are unclear. The mechanisms of this a matter of speculation, but some possibilities include reductions in oral nutritional intake that accompany development of GVHD, changes in dietary behavior or preferences by patients, effects of oral medications, disruption of the production of antimicrobial peptides by intestinal epithelial cells, changes in bile acid metabolism or secretion, or changes in gut transit time in the setting of diarrhea.

A potential consequence of acute GVHD-associated changes in intestinal bacterial composition could be alterations in the metabolites produced by these bacteria. Interestingly, in accordance with this hypothesis, a recent study reported that unbiased profiling of bacterial metabolites identified butyrate as significantly depleted in the intestinal tissues of mice with acute GVHD.⁹¹ Butyrate is a short-chain fatty acid produced by members of the intestinal microbiota and is known to induce regulatory T cells.^{92,93} Butyrate can inhibit histone deacetylases, and intestinal epithelial cells exhibited reduced histone acetylation in the setting of acute GVHD. Oral feeding with butyrate or, alternatively, introduction of a mixture of butyrate-producing Clostridia led to restoration of intestinal epithelial histone acetylation and improved survival in mice with acute GVHD. Clinical strategies to increase intestinal butyrate levels are in active development. For example, preclinical evidence suggests potato-based starches can induce production of butyrate from the microbiota,⁹⁴ and a randomized trial of this prebiotic for acute GVHD prevention is underway (www.clinicaltrials.gov: NCT02763033).

Separate from the development of acute GVHD in which alloreactive T cells mount an immune response against host tissues, other clinical developments during HCT can lead to major changes in intestinal bacteria.⁹⁵ Administration of antibiotics during the HCT hospitalization explains the majority of these changes, frequently in the form of prophylactic antibiotics or empiric antibiotics for neutropenic fever. The choice of antibiotic can determine the degree of microbiota disruption, and agents with more limited spectra of activity appear to be associated with reduced acute GVHD severity.^{96,97} For example, rifaximin was recently reported to perform favorably as bacterial prophylaxis in comparison with a historical cohort treated with a prophylactic combination of ciprofloxacin and metronidazole.⁹⁷ Rifaximin use was associated with both better preservation of intestinal commensal bacteria and reduced transplant-related mortality. Similarly, antibiotics used to treat neutropenic fever that spare obligate anaerobes, including cefepime and aztreonam, produce less perturbation to intestinal bacterial composition compared with broader-spectrum antibiotics with increased anaerobic activity such as piperacillin-tazobactam and carbapenems.⁹⁶ Use of narrow-spectrum antibiotics was associated with reduced acute GVHD lethality in humans and reduced acute GVHD severity in animal models. Prospective studies exploring different antibiotic regimens for prevention or treatment of neutropenic infections while minimizing intestinal

bacterial repercussions are now beginning to be investigated (www.clinicaltrials.gov: NCT02641236).

How intestinal bacteria can modulate acute GVHD severity remains an open question. As noted earlier, early murine studies from the 1970s with mice transplanted in germ-free conditions¹ or while receiving gut-decontaminating antibiotics,² as well as clinical studies based on these findings,⁹⁸⁻¹⁰⁰ suggested that intestinal microbes, as a whole, contribute to development of acute GVHD. Potentially proinflammatory commensal bacteria include Enterobacteriales and *Enterococcus* in both murine^{65,64} and clinical^{101,102} acute GVHD. Other intestinal bacterial populations, however, appear to help reduce acute GVHD, including *Lactobacillus*^{64,103} and Clostridia.^{91,104-106} Thus far, identification of specific individual bacteria, or consortia of bacteria, that are important for ameliorating GVHD has not been accomplished but is an active area of investigation. Concurrently, clinical trials in which patients receiving allogeneic HCT are being treated with introduction of a complete intestinal microbiome are actively enrolling. Strategies include autologous fecal microbiota transplantation using a patient's own personal microbiome that was collected and preserved pre-HCT (www.clinicaltrials.gov: NCT02269150), as well as introduction of a third-party intestinal microbiome donated by a healthy volunteer that can be introduced to patients in the form of capsules (www.clinicaltrials.gov: NCT02733744). Notably, both of these approaches have already been shown to be beneficial for the treatment of *Clostridium difficile* infection, a disease that is closely associated with antibiotic-related intestinal microbiota disruption.^{107,108}

A growing body of literature has highlighted the importance of the mucus layer as a critical line of defense in the intestinal mucosa.¹⁰⁹ Produced and secreted by goblet cells, mucus forms a molecular network that concentrates epithelial-derived antimicrobial molecules in the intestinal lumen adjacent to the epithelium and, particularly in the colon, also generates a dense physical barrier that is difficult for most intestinal bacteria to penetrate.¹¹⁰ The building block of intestinal mucus is the glycoprotein *MUC2*, a rich source of carbohydrates. *MUC2* is continuously secreted by goblet cells as well as consumed by intestinal bacteria, which harbor enzymes that allow metabolism of *MUC2* glycans. Loss of the mucus layer was recently observed in mice with acute GVHD aggravated by broad-spectrum antibiotics, indicating that changes in intestinal bacterial composition can have an effect on the health and integrity of the mucus layer.⁹⁶ Clinical evidence suggests the mucus layer can, in turn, shape the composition of the intestinal microbiota. Fucosylation of mucus glycans can vary between individuals, which is predominately determined by the function of the fucosyltransferase *FUT2*. Polymorphisms of the *FUT2* gene are common and result in 2 different phenotypes, including so-called secretors and nonsecretors. Interestingly, nonsecretors have been found to harbor intestinal microbiota with reduced species richness,¹¹¹ and nonsecretor status in a study of 150 patients was found to be associated with a decreased risk for acute GVHD, but an increased risk for bacteremia after allogeneic HSCT.¹¹²

Interactions between bacterial and fungal commensals within the intestinal tract are also beginning to be uncovered. Commensal bacteria, including Clostridia and *Bacteroides*, are important for preventing overgrowth of *Candida* in the intestinal tract.¹¹³ Heat-killed *Candida albicans*, as well as purified α -mannan, a major component of fungal cell wall, can aggravate acute GVHD in mouse models.¹¹⁴ In addition, pharmacologic inhibition of *Candida* in the

form of fluconazole was found to reduce acute GVHD in a randomized, placebo-controlled study of prophylaxis of invasive candidiasis.¹¹⁵ *Candida* colonization has also been found to be associated with increased acute GVHD, although the interaction may have an extra layer of complexity, as the association was only observed in the subset of patients with a wild-type dectin-1 genotype. Patients lacking wild-type dectin-1, an innate pattern receptor that can recognize *Candida*, did not show an association between candida colonization and acute GVHD.¹¹⁶

Finally, a potential link between enteric viruses and acute intestinal GVHD has been recognized for some time. CMV reactivation after allogeneic HCT has been found to be associated with an increased risk of developing acute GVHD,¹¹⁷ raising the possibility that CMV replication may contribute to allogeneic T-cell activation. This may also be true for other viruses with potential tropism for intestinal epithelium; a recent report found that, using a panel of PCR assays against enteric viruses, intestinal virus positivity before allogeneic HCT was associated with later development of acute GVHD.¹⁴ Interestingly, the vast majority of virions in the intestinal tract are nonpathogenic bacteriophages that can only infect and replicate within bacteria. Specific changes, however, in the profile of these bacteriophages have been observed in patients with ulcerative colitis, which in turn are distinct from those found in patients with Crohn's disease.¹¹⁸ This raises the possibility that perturbations in both the nonpathogenic and pathogenic subsets of the intestinal virome may also be linked with intestinal GVHD.

Future strategies

Recent work has resulted in a renewed appreciation for the bidirectional relationship that exists between the intestinal mucosa and intestinal commensals, which can directly have an effect on the development and maintenance of acute intestinal GVHD. As new tools to characterize the intestinal mucosa and the intestinal microbiota become available, we will come closer to a time when manipulation of the interactions between intestinal microbiota, host tissues, and immune function can be incorporated into the care of patients.

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